Remote Sensing

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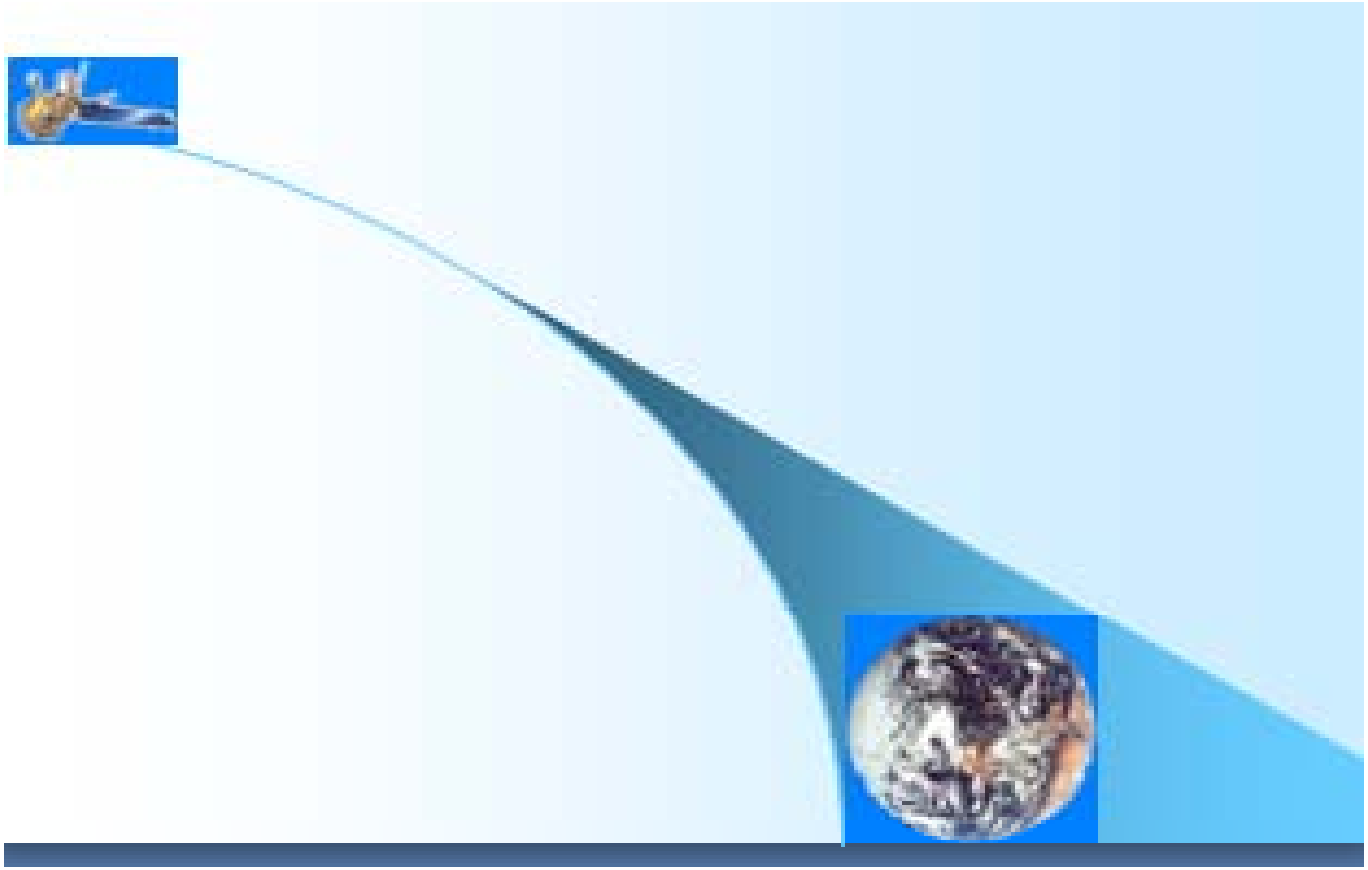
WHAT IS REMOTE SENSING

The act of recording / observing / perceiving (**Sensing**) objects or events at far away (**Remote**) places.



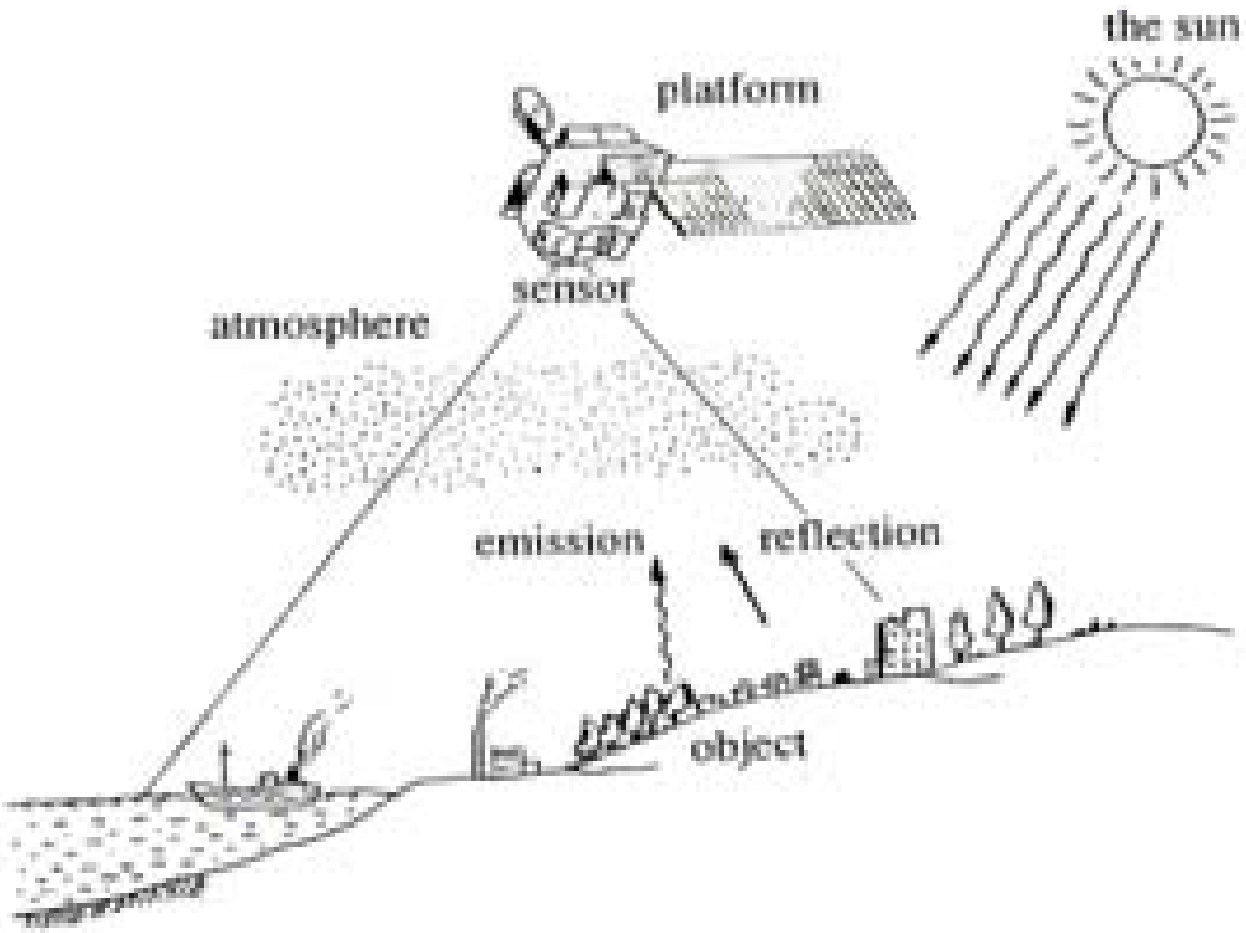
WHAT IS REMOTE SENSING

•The science and art of acquiring information (Spectral, Spatial, Temporal) about our surrounding (Objects, Area, Phenomenon).

•No physical contact between Sensor and Objects.
We do Remote Sensing all the time!!!

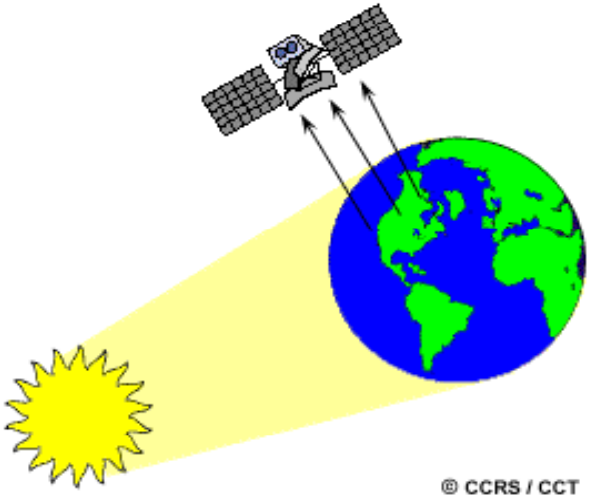
•Remote Sensing is the science and art of obtaining information about the object, area or phenomenon through the analysis of data acquired by a device that is not in contact with the object, area or phenomenon under investigation (Lillesand and Kieffer 1987).

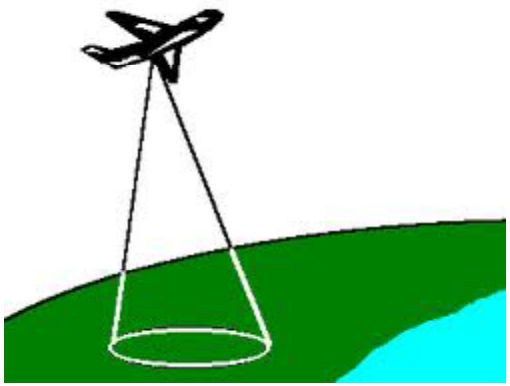
Components of Remote Sensing

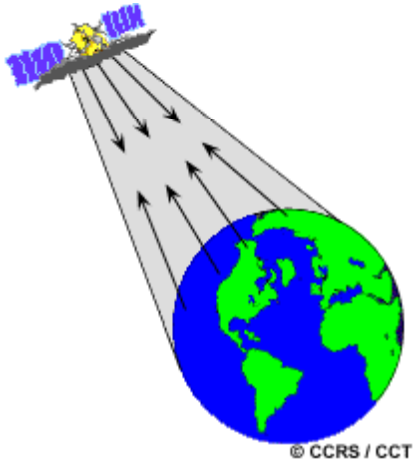


Types of Remote Sensing

Passive vs. Active Sensing







AN IDEAL REMOTE SENSING SYSTEM

System component The 'ideal' system

- **A uniform energy source**
- **A non-interfering atmosphere**
- **A series of unique energy / matter interactions at the Earth's surface**
- **A super sensor**
- **A real-time data handling system**
- **System component The 'ideal' system Multiple data users**

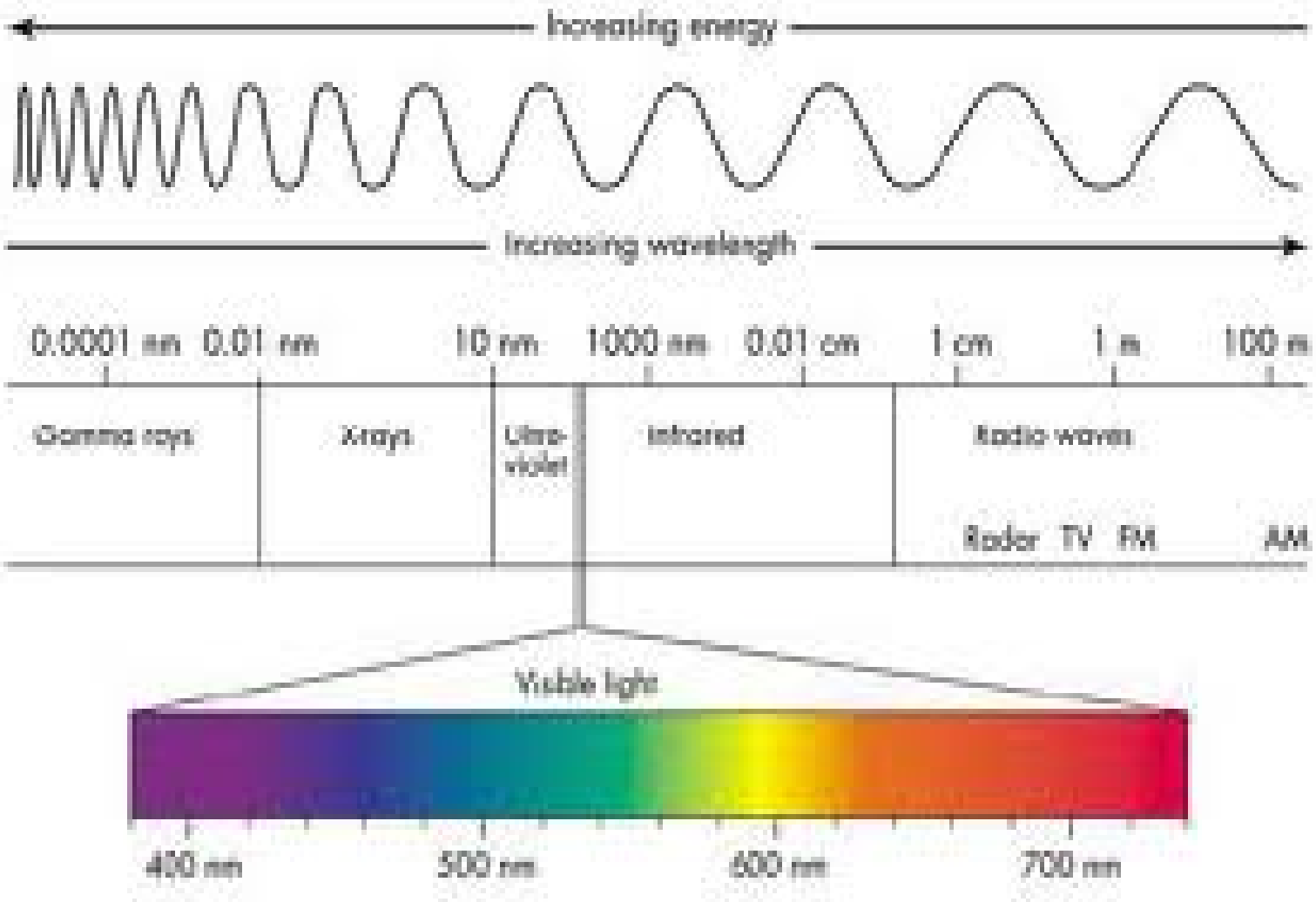
An Real remote sensing system

- The energy source
- The atmosphere
- The energy / matter interactions at the Earth's surface.
- The sensor.
- The choice of a sensor for any given task always involves tradeoffs
- The data-handling system
- The multiple data users.

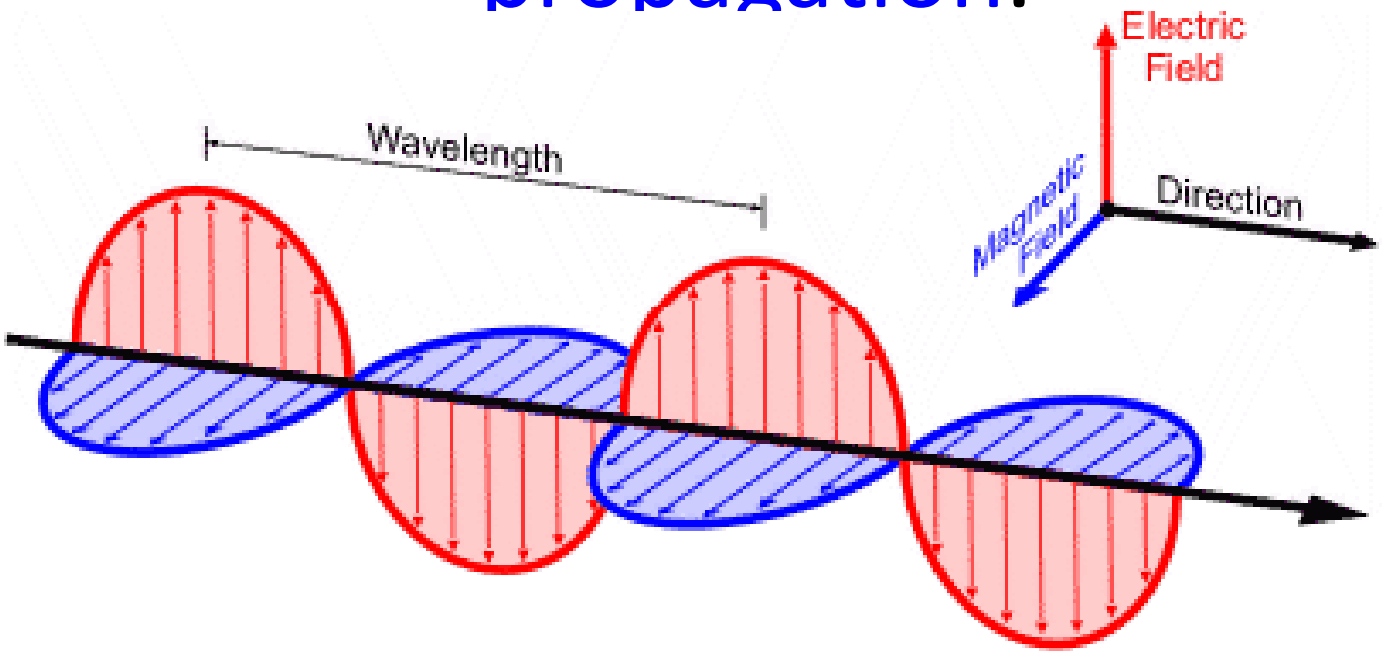
Fundamentals Concepts of Remote Sensing

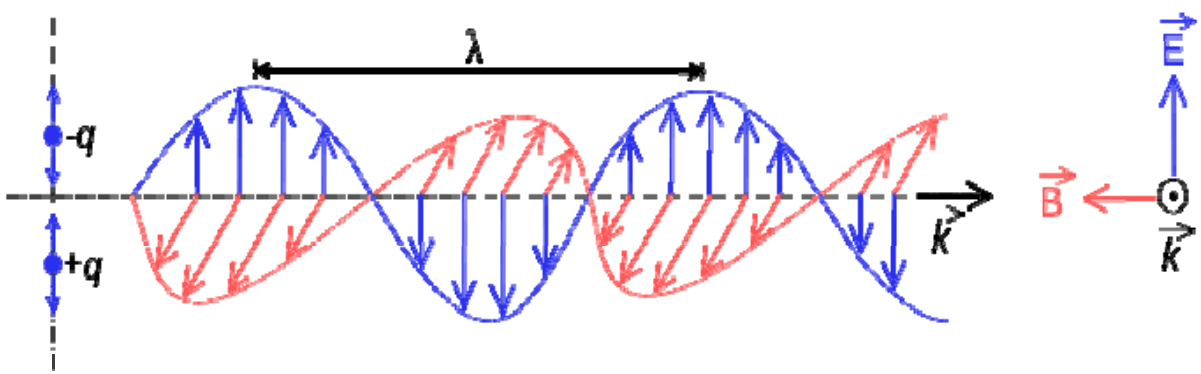
The electromagnatic spectrum (EMR), Remote Sensing in visible IR spectrum and thermal IR spectrum

Energy source and reflection principles
– Wave theory
– Particle theory

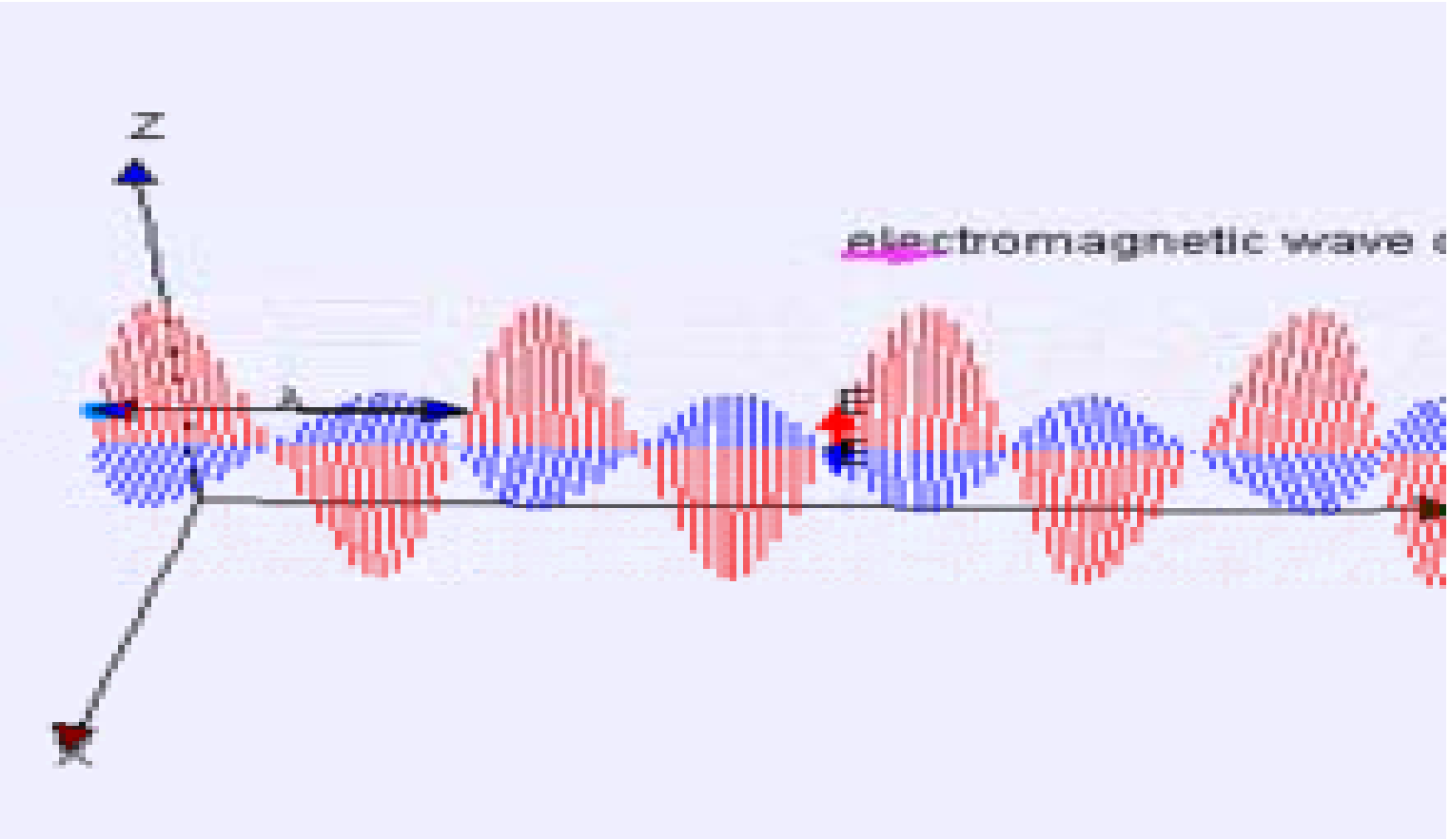

Increasing energy
Increasing wavelength
0.0001 nm
0.01 nm
10 nm
1000 nm
0.01 cm
1 cm
1 m
100 m
Gamma rays
Xrays
Ultra-violet
Infrared
Radio waves
Radar
TV
FM
AM
Visible light
400 nm
500 nm
600 nm
700 nm

Direction of energy and wave propagation.





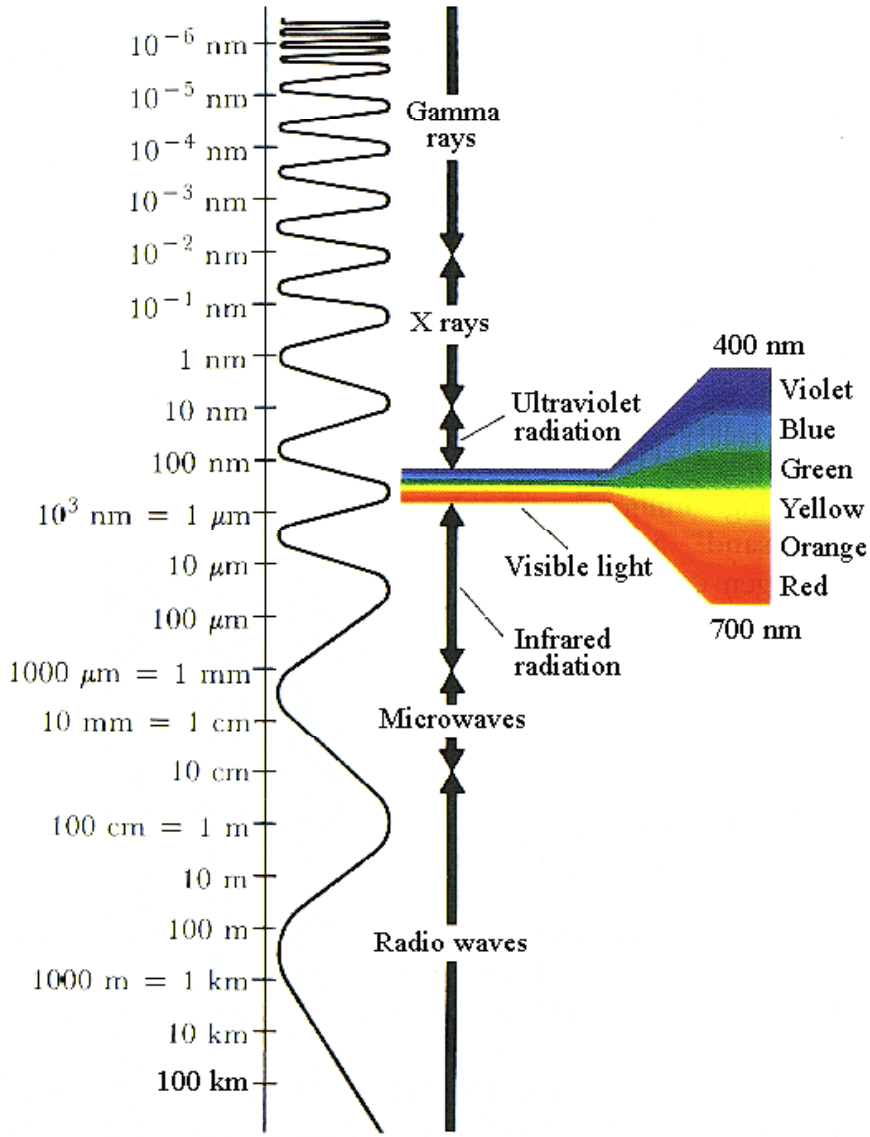
Direction of energy and wave propagation



Visible to IR spectrum

the frequencies in this portion of the spectrum are in the millions of megahertz, it is customary to refer to wavelength rather than frequency when describing them. The unit most commonly used to describe wavelengths of visible and infrared radiation is the micron (μm or) which is 10^{-6} meters long. Visible light extends from approximately 0.4 to 0.76 microns and is broken down into bands according to color. Infrared (IR) extends from 0.76 to 1000 microns and is divided into bands called near (NIR), middle (MIR), far (FIR), and extreme (XIR) infrared. There are, in fact, no clear-cut boundaries to the various portions of the electromagnetic spectrum. The limits used are somewhat arbitrary, and may vary from publication to publication.

Wave length



Wave theory: $c = nl$

c : speed of light (3×10^8 m/s)

n : frequency (cycle per second, Hz)

l : wavelength (m)

unit: micrometer mm = 10^{-6} m

Spectrum :

UV (ultra<u>violet</u>)

Vis (visible)

narrow range, strongest, most sensitive to human eyes

blue: 0.4~0.5mm

green: 0.5~0.6mm

red: 0.6~0.7mm

IR (infra<u>red</u>)

near-IR: 0.7~1.3 mm

mid-IP: 1.3~3.0 mm

thermal-IR: 3.0 mm~1mm ← heat sensation

microwave: 1mm~1m

Spectral Reflectance

$$E_I(\lambda) = E_R(\lambda) + E_T(\lambda + E_A(\lambda)$$

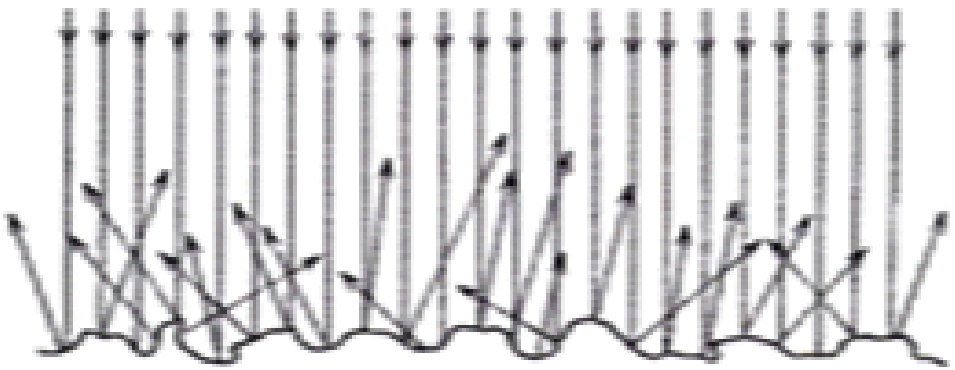
$$E_R(\lambda) = E_I(\lambda) - E_A(\lambda) + E_T(\lambda)$$

Spectral Reflectance is ρ

Therefore

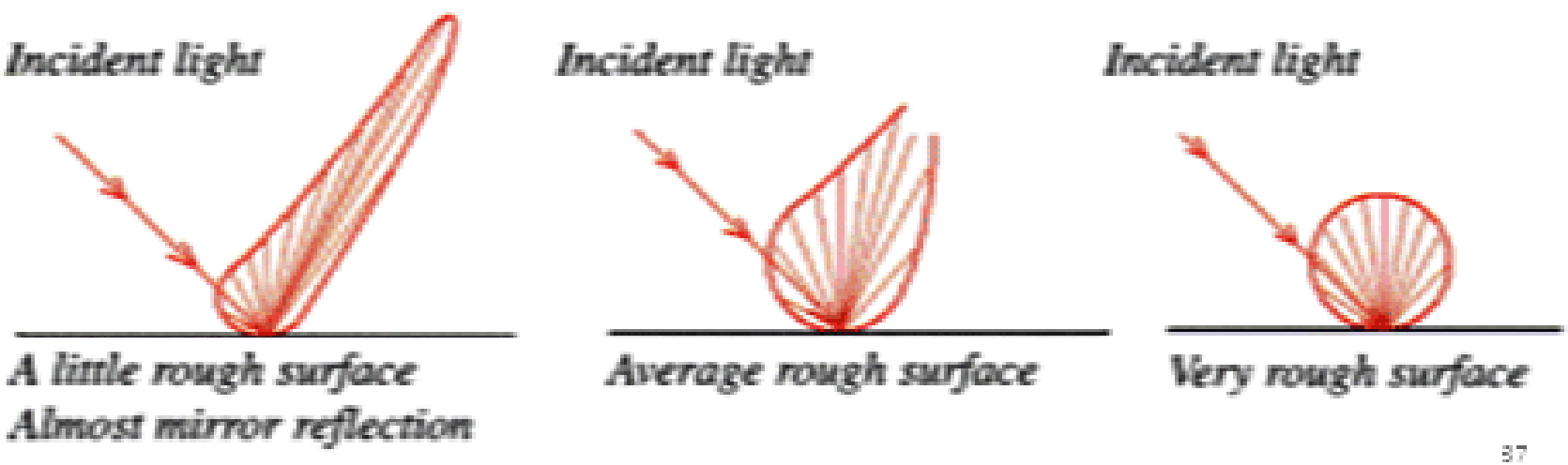
$$\rho\lambda = \frac{\text{Energy of wavelength } \lambda \text{ reflected from the object}}{\text{Energy of wavelength } \lambda \text{ Incident upon the object}} \times 100$$

Reflectance

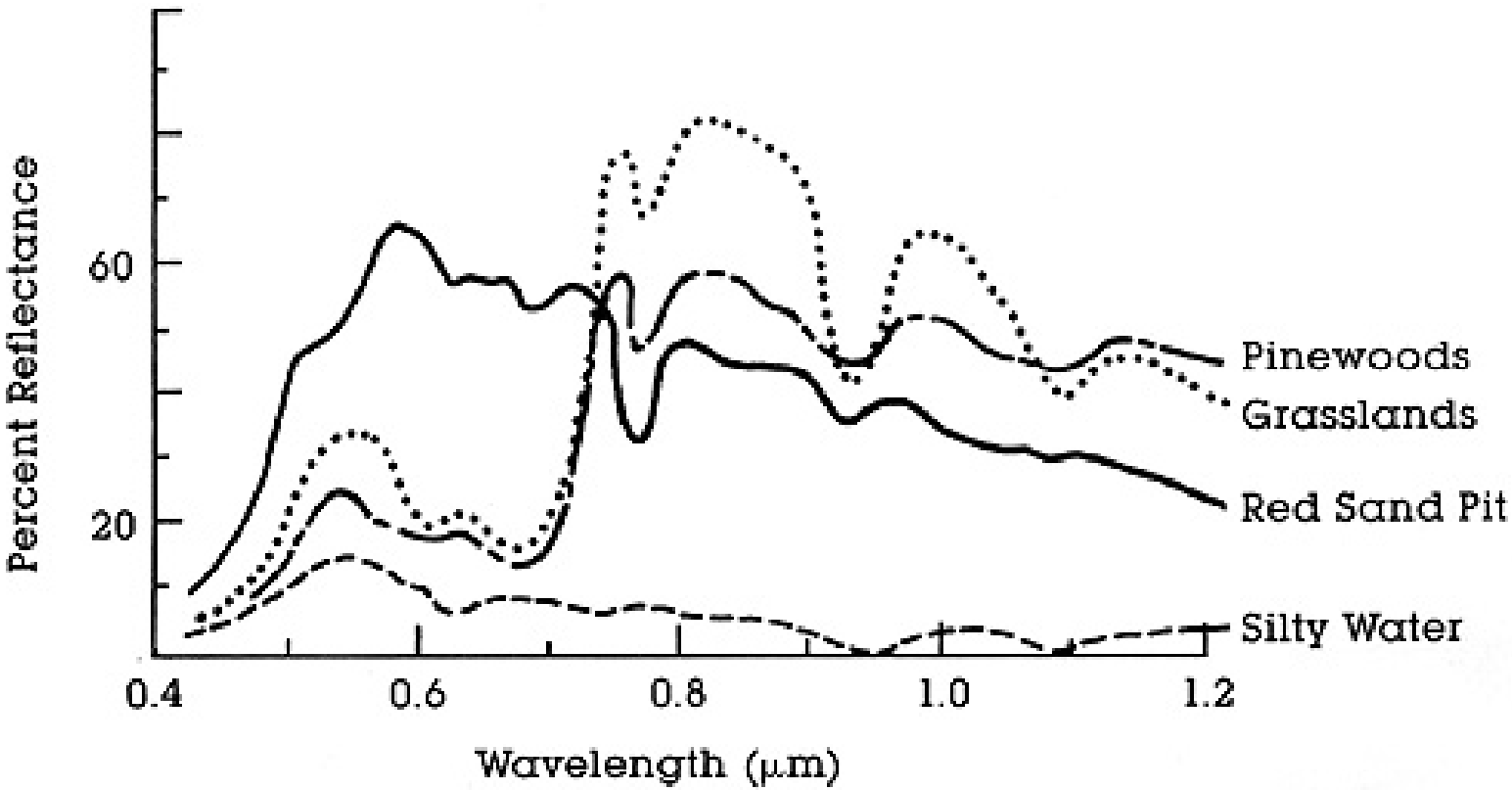


Light reflection on a irregular surface

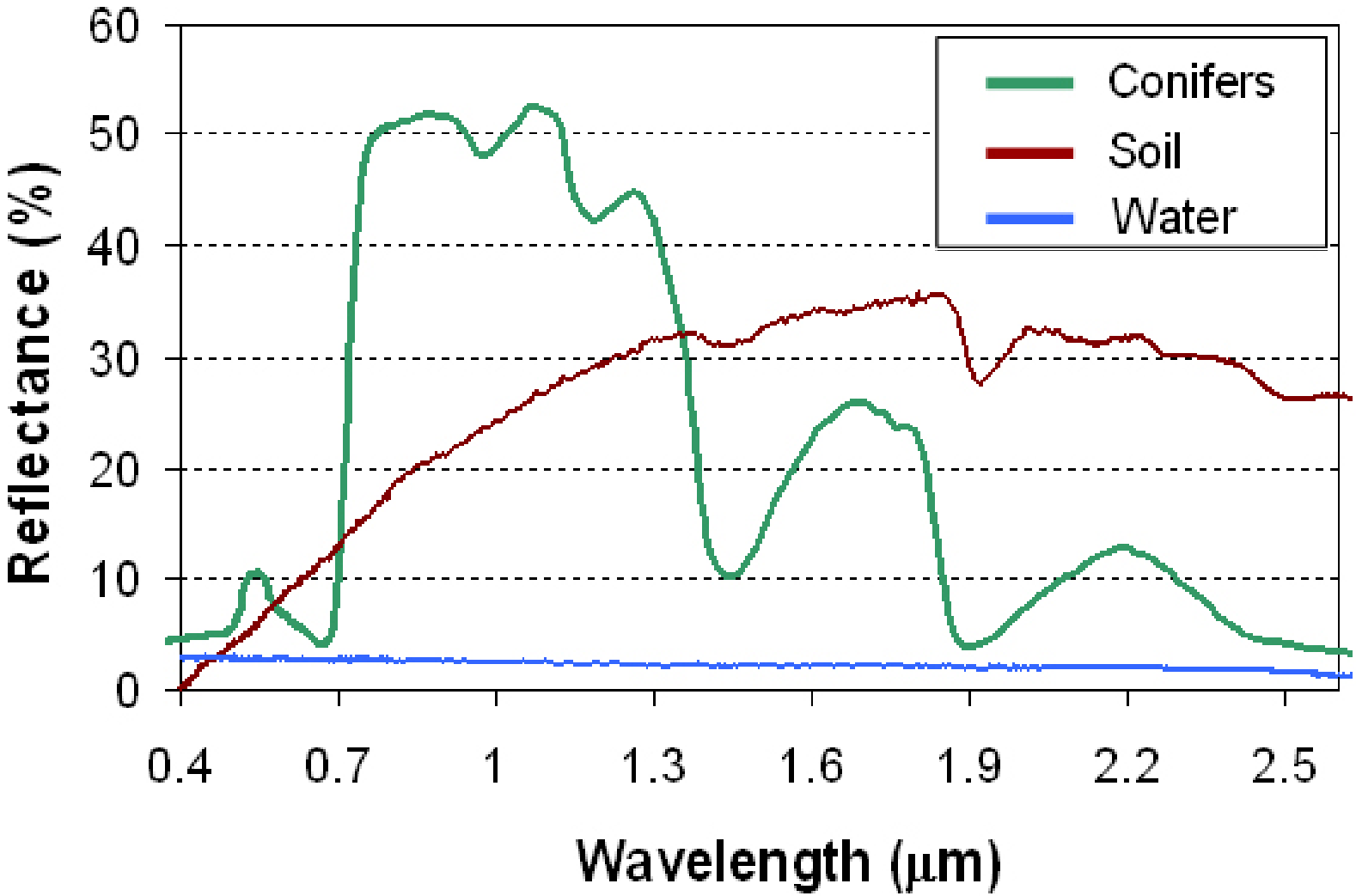
- Specular
- Diffuse



Spectral reflectance



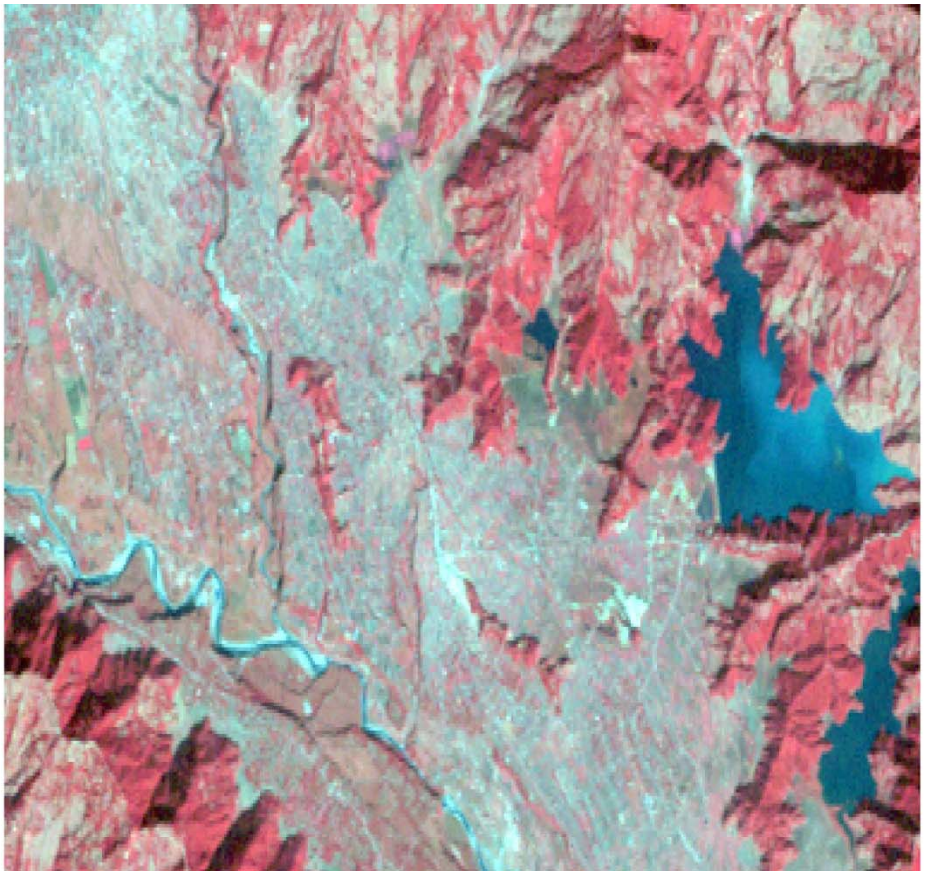
Spectral signature of an object



Spectral signature

The figure implies that the mentioned feature types are normally spectrally separable. The degree of separation of landscape features depends, however, on the range of wavelengths we are considering. This means that two features may be indistinguishable for a certain spectral range and perfectly individualized for other spectral band.

Spectral signatures

Spectral signatures are the specific combination of emitted, reflected or absorbed electromagnetic radiation (EM) at varying wavelengths which can uniquely identify an object

Landsat image